Chapter 1

Latent Space Analysis

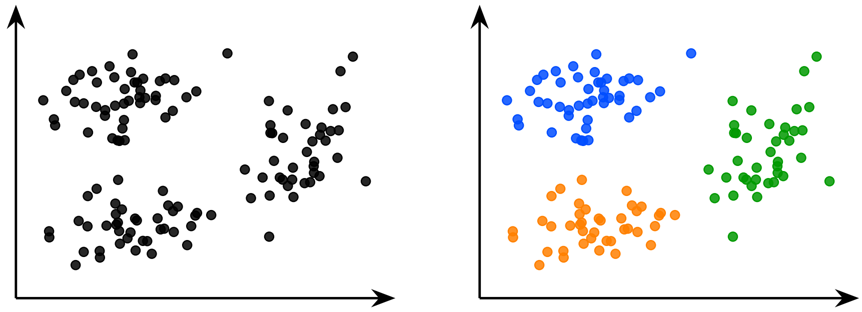
The ability of a neural network to generalize provides evidence that the model has learned more than a simple memorization of the training examples. Instead, it must construct internal representations that capture regularities present in the data. These representations play a central role in the network's ability to solve a task. Understanding their structure is therefore a key step toward understanding how neural networks achieve generalization.

In classical machine learning approaches, practitioners had to manually design data representations in order to make them usable by simple models. Deep learning introduces a paradigm shift: rather than relying on hand-crafted features to meaningfully represent data, it automatically learns these representations. Through the successive activation of hidden layers, neural networks gradually construct representations that become increasingly relevant to the task under consideration. As discussed in Chapter **??**, these representations are not meaningful at initialization, they emerge progressively during training as the model optimizes its objective. It is precisely through these learned representations that a trained network is able to map an input to an output, including for examples never encountered during training.

The study of these representations has therefore become a central topic in deep learning research, particularly in understanding how information is organized within them, what concepts they encode, and how they influence model behavior. In this chapter, we first make explicit the working hypothesis that underlies most research on latent space analysis. In this manuscript, we refer to this assumption as the "concept hypothesis." To do so, we introduce the notions of abstraction, concept, latent space, and semantics. Once the notions of this conceptual framework has been established, we review the main methods that aim to identify and extract concepts from latent representations.

1.1 Concept Hypothesis

In this section, we introduce the main working hypothesis underlying much of the literature on latent space analysis. Although rarely stated explicitly, this hypothesis is implicitly present in the recurring use of notions such as abstraction, representation, latent space, concept, and semantics. The goal of this section is to make this conceptual framework explicit by formally defining its main



(a) Observed activations **(b)** Under the concept hypothesis

Figure 1.1: The same latent space activations seen without concept hypothesis **(a)** and under the working hypothesis that concepts emerge in the latent space **(b)**. Colors represent hypothetical concept memberships assigned by the model; they are not observed labels. Under this hypothesis the structure of the latent space becomes meaningful and amenable to analysis.

components.

This hypothesis originates from the need to explain the generalization ability of deep neural networks observed after training, as discussed in Section **??**. Formally, a model is said to generalize when, for an input $x \in \mathcal{X}$ not seen during training, it still produces an output $\hat{y}$ close to the true output y. Such behavior cannot be explained by simple memorization of training samples. Instead, the model must extract and exploit underlying regularities in the target function. This requires a process of abstraction, as defined in Definition 1.1.1, whereby raw inputs are transformed into internal structures that are meaningful for the task.

Definition 1.1.1: Abstraction

Abstraction is the process by which a model transforms an input into a task-relevant internal form. It captures and restructures information in order to represent relationships that are useful for prediction or decision-making.

Through this abstraction process, a neural network does not directly map x to y. Instead, as illustrated in Figure **??**, it computes a sequence of intermediate activations across hidden layers. These activations correspond to internal encodings of the input, which we refer to as representations in the sense of Definition 1.1.2.

Definition 1.1.2: Representation

A representation is the internal encoding of an input produced by a trained neural network. It corresponds to the activations of one or several layers and provides a transformed view of the input.

Each hidden layer refines the representation of the input. It progressively transforms it into a structure more directly aligned with the output space. This process can be described as a sequence of transformations where each $h^{(l)}$ is a representation:

$$x \to h^{(1)} \to h^{(2)} \to \cdots \to h^{(L-1)} \to y$$

The set of all such representations produced by a layer or a model over a dataset defines its latent space, as defined in Definition 1.1.3.

Definition 1.1.3: Latent Space

The latent space is the set of all representations produced by a layer or model over a dataset. It is shaped by learning and organizes representations according to task-relevant properties.

The existence of meaningful structure in the latent space is closely related to generalization. Under the concept hypothesis, we assume that the neural network internally forms task-dependent abstractions, hereafter referred to as concepts and formally defined in Definition 1.1.4. These concepts are not directly observable, but are expected to manifest in the latent space as recurring and structured patterns in the learned representations. Under this view, a representation can be interpreted as a composition of multiple concepts, each contributing to the model's prediction.

Definition 1.1.4: Concept

A concept is a task-dependent abstraction that captures regularities in the input space relevant for predicting the target output.

Another property that can be considered when analyzing concepts is their relationship to human interpretation. This relationship is captured by the notion of semantics, defined in Definition 1.1.5. Under this definition, concepts may exhibit varying degrees of semantic content. Some concepts can be readily associated with human-interpretable notions, whereas others may contribute to task performance while remaining difficult to interpret.

Definition 1.1.5: Semantics

Semantics refers to the alignment between concepts in the latent space and concepts used by a human observer. A structure is semantic if it can be consistently associated with a human-interpretable concept.

It is important to note that this conceptual framework is not yet supported by a complete theoretical understanding of deep neural networks. Current mathematical tools do not provide a rigorous characterization of their internal representations and do not formally prove that concepts emerge as identifiable structures within latent spaces. Nevertheless, this perspective constitutes a widely adopted working hypothesis in the field of representation and latent space analysis.

The remainder of this chapter is devoted to reviewing methods for analyzing latent spaces and extracting the concepts they encode.

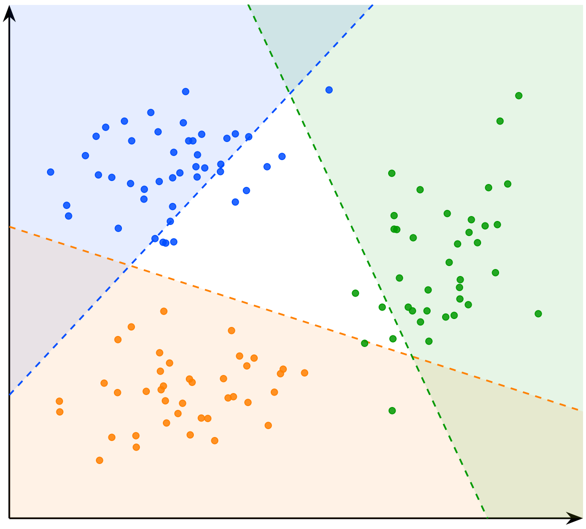


Figure 1.2: Decision boundaries of the linear probes for concepts A, B, and C. The colored regions indicate the areas classified positively by each probe.

1.2 Probing

Probing methods aim to determine whether a given human-defined concept is encoded within the internal representations of a neural network. More specifically, they investigate whether the presence or absence of a selected concept can be inferred from embeddings. The underlying assumption is that if a concept can be reliably decoded from these representations, then the network likely encodes information related to it in order to perform its task.

A probing analysis begins by identifying a set of interpretable concepts that the network could potentially use to accomplish its task. Each example in the dataset is then annotated to indicate the presence or absence of the studied concepts. We denote by c_j the presence or absence of concept j in example x. The dataset can therefore be written as follows:

$$\mathcal{D} = \{(x^{(i)}, y^{(i)}, c_1^{(i)}, \ldots, c_n^{(i)})\}_{i=1}^{N}.$$

The data are subsequently projected into the latent space of the pretrained model. We denote by h^l the representation vector obtained at layer l. From these latent representations, a classifier is trained in a supervised manner to predict the presence of the considered concept. This classifier is referred to as a probe, denoted p, since its role is to determine whether a concept is present or not. Its purpose is therefore to assess whether information associated with the concept is accessible from the network's internal representations. This can be formalized as follows:

$$p_\theta^{l,j} : h^l \mapsto c_j, \quad \theta^* = \arg\min_\theta \mathcal{L}\big(p_\theta^{l,j}(h^l), c_j\big)$$

The classifier used to detect the presence or absence of a concept must remain intentionally simple. Indeed, if the probe has too much capacity, it may reconstruct the target information from complex correlations that do not necessarily correspond to information explicitly encoded in the latent space. In such a case, it becomes difficult to determine whether the concept is genuinely represented by the network or merely reconstructed through the expressive power of the classifier. For this reason, probing methods often rely on linear classifiers. A linear separation suggests that the information is easily accessible to the neural network and potentially usable during decision-making. If the classifier achieves performance significantly above chance level, this suggests that the latent space contains exploitable information related to the considered concept.

An application of this technique can be found in studies investigating whether large language models (LLMs) develop internal representations of their environment [**?**, **?**]. In these works, researchers train LLMs to play games such as chess or Othello in a purely textual setting. The model is only provided with sequences of moves and information about the actions it can take, all expressed in natural language. The central question is whether the LLM develops an internal representation of the game state. To address this question, the probed concepts correspond to the presence or absence of different game pieces on specific board positions. What these studies observe is that probes are able to reconstruct the full board state from the model's hidden representations, even though the only available information to the model is a sequence of text.

However, the fact that a concept can be decoded from latent representations does not necessarily imply that the model actually uses this information to perform its task. To assess whether representations play a causal role in the model's output, it is possible to perform interventions on the latent representations. An intervention consists of modifying a latent representation in a controlled manner by activating or deactivating a given concept, and then studying its impact on the model's output. If the resulting change in the model's output is consistent with the applied perturbation, this suggests that the representation had a causal role.

To this end, we aim to minimally modify the latent representation h^l such that the probe prediction for the target concept c_j changes state, while leaving the predictions associated with other concepts unchanged. This leads to an optimization problem that can be formulated as follows. A common approach to solve this problem is to perform gradient-based optimization on the latent activation h^l, in order to directly optimize the target objective:

$$h^{l*} = \arg\min_{h^{l'}} \mathcal{L}_{\text{flip}}\big(p_\theta^{l,j}(h^{l'}), c_j\big) + \lambda \sum_{k \neq j} \mathcal{L}_{\text{preserve}}\big(p_\theta^{l,k}(h^{l'}), c_k\big) + \beta \|h^{l'} - h^l\|_2^2$$

where:

- $\mathcal{L}_{\text{flip}}$: loss encouraging the prediction of the target concept c_j to change state (i.e., flipping the predicted label),
- $\mathcal{L}_{\text{preserve}}$: loss enforcing invariance of all non-target concepts c_k for $k \neq j$,
- $\|h^{l'} - h^l\|_2^2$: regularization term ensuring the modified representation remains close to the original latent representation,
- λ, β: weighting coefficients controlling the trade-off between preservation, intervention strength, and minimality of the perturbation.

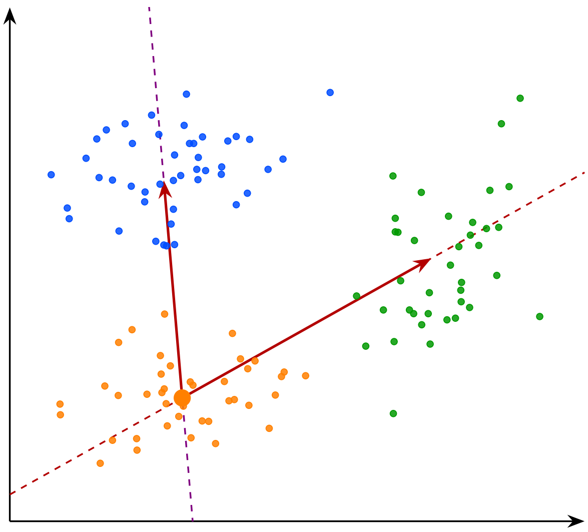


Figure 1.3: Directions interprétables dans l'espace latent reliant le cluster B aux clusters A et C. Les lignes pointillées représentent les directions globales, tandis que les flèches indiquent les vecteurs entre centroïdes.

In the context of LLMs studied previously, such interventions have been used to remove specific pieces from the model's internal representation of the game state [**?**, **?**]. The main observation is that, in the vast majority of cases, the model behaves as if the piece had effectively disappeared. In other words, the LLM continues to generate legal moves while correctly accounting for the absence of the removed piece. This provides evidence for the causal role of these internal representations in the model's behavior.

A main limitation of the probing approach is that the concepts under investigation must be predefined before any analysis can be performed. This introduces a strong bias in what can be studied. Moreover, annotating a dataset for each concept is extremely time-consuming and requires significant human effort. Interventions in latent space are also difficult to interpret, as it is not always clear what constitutes a meaningful change in the network's output under a given perturbation.

Probing methods provide a binary view of concepts, focusing on whether a given concept is present or absent in a latent representation. However, this binary perspective can be overly restrictive, as concepts may be represented in a more continuous or graded manner, with varying degrees of presence. To obtain a more detailed characterization of these representations, other approaches have been proposed, such as the analysis of interpretable directions in latent spaces.

1.3 Interpretable Directions

In latent space analysis, probing methods are generally limited to a relatively simple task: predicting the presence or absence of a given concept from a latent representation. This approach makes it possible to assess whether information is encoded in the latent space, but it does not provide a detailed characterization of its structure.

A more advanced line of investigation focuses on directions in latent space. Let $\mathbf{z} \in \mathbb{R}^d$ denote a latent representation. Empirically, in certain models, moving a latent representation along a direction $\mathbf{v} \in \mathbb{R}^d$ results in coherent changes in the model's output.

More precisely, consider the perturbed representation

$$\mathbf{z}' = \mathbf{z} + \alpha \mathbf{v},$$

where $\alpha \in \mathbb{R}$ controls the magnitude of the intervention. The resulting change in output often varies smoothly with α, suggesting an approximately linear structure in latent space with respect to certain semantic factors. This observation has led to the hypothesis that neural networks may organize representations in a partially linear manner with respect to semantic concepts.

Such vectors $\mathbf{v}$ are referred to as <u>interpretable directions</u>, since they induce transformations that can be associated with semantic meaning. The goal of latent space analysis through interpretable directions is therefore to identify these semantically meaningful axes of variation.

This phenomenon has been extensively studied in generative image models. In this context, controlled perturbations of latent representations along interpretable directions can induce specific semantic transformations, such as object rotation, scaling, or age modification in facial images. These observations have led to the development of the field of latent space editing, whose objective is to manipulate generated content directly through modifications in latent space rather than through changes in the input prompt.

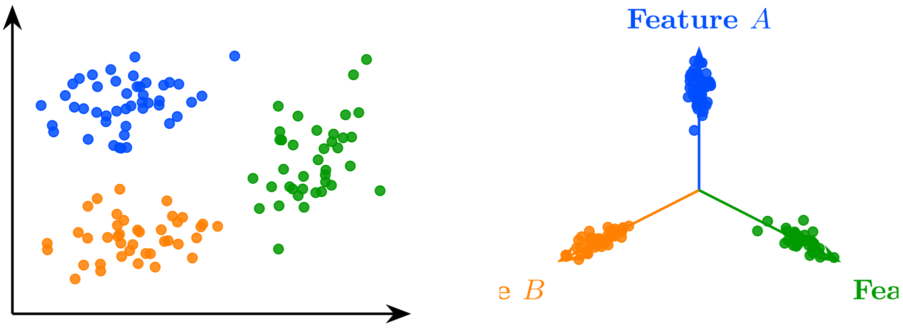
However, a central challenge remains: identifying such interpretable directions is far from trivial. Most existing approaches rely on auxiliary models that encode human-defined semantics, such as CLIP, to relate latent variations to textual concepts. Consequently, these methods inherit a limitation similar to that of probing techniques, since concept discovery depends on a pre-existing semantic framework defined by humans.

To overcome this limitation, a research direction known as unsupervised semantic discovery has emerged. Its goal is to uncover interpretable directions in latent spaces without explicit semantic supervision. In these approaches, one model applies perturbations along random directions in latent space, while a second model attempts to predict or recognize the applied transformation. If a stable correspondence emerges between the perturbation and its prediction, this suggests that the considered direction corresponds to a coherent structure in latent space.

In this framework, human intervention occurs only a posteriori, to verify whether the discovered directions indeed correspond to interpretable semantic variations. This approach enables the capture of representations that go beyond the mere presence or absence of a concept, including notions such as the intensity or degree of an attribute. As such, it follows a philosophy similar to causal probing, while also enabling explicit interventions to assess the causal role of latent representations.

However, despite these advances, several challenges remain.

First, it appears that not all concepts encoded by neural networks are linearly identifiable in latent space. While some directions correspond to well-defined semantic factors, many others do not admit such a simple linear structure.



(a) Dense latent space **(b)** Sparse SAE space

Figure 1.4: Transformation de l'espace latent dense vers une représentation sparse de type SAE. À gauche **(a)**, les embeddings sont entremêlés dans l'espace latent initial. À droite **(b)**, la représentation sparse aligne chaque concept sur un axe de feature dédié.

In addition, latent representations often exhibit a phenomenon known as entanglement. Let $f : \mathbb{R}^d \to \mathbb{R}^k$ denote a mapping from latent space to a set of semantic attributes. Ideally, one would like to associate each semantic factor $i \in \{1, \dots, k\}$ with a direction $\mathbf{v}_i \in \mathbb{R}^d$ such that:

$$f(\mathbf{z} + \alpha \mathbf{v}_i)_j \approx f(\mathbf{z})_j \quad \text{for } j \neq i,$$

and

$$f(\mathbf{z} + \alpha \mathbf{v}_i)_i \text{ varies significantly with } \alpha.$$

In practice, this property does not hold strictly. Instead, a single direction $\mathbf{v}$ often induces changes in multiple components of $f(\mathbf{z})$, i.e.:

$$\frac{\partial f_j(\mathbf{z})}{\partial \alpha} \neq 0 \quad \text{for several } j.$$

This phenomenon is referred to as entanglement, indicating that semantic attributes are not cleanly separable in latent space.

This difficulty in isolating independent semantic factors, together with the frequent presence of entangled representations, has motivated a more ambitious hypothesis: the superposition hypothesis.

1.4 Sparse Autoencoder

As discussed previously, interpretability approaches based on latent directions face inherent limitations. Empirically, only a restricted subset of concepte appears to admit a linear representation, and even these directions are often affected by significant entanglement with other concepte.

To account for this behavior, a complementary hypothesis has been introduced, extending the assumption of linear concepte encoding in neural networks. It posits that semantic concepte are preferentially represented in a linear fashion, with each concept ideally corresponding to a distinct direction in latent space.

In this idealized setting, semantic attributes would form an approximately orthogonal basis, where each direction independently controls a single factor of variation.

We first consider an idealized representation of the latent space in which semantic concepts are perfectly disentangled. Let $\mathbf{h} \in \mathbb{R}^d$ denote a latent embedding. In this setting, each concept is associated with a unique direction $\mathbf{v}_i \in \mathbb{R}^d$, and the representation of a given input can be expressed as a linear combination of these concept directions:

$$\mathbf{h} = \sum_{i=1}^{m} a_i \mathbf{v}_i,$$

where $a_i \in \mathbb{R}$ represents the activation of concept i.

In the ideal case, the set of concept directions is assumed to be approximately orthogonal, such that each direction independently controls a single semantic factor. Moreover, for a given input, only a small subset of coefficients a_i is non-zero, so that each embedding corresponds to a sparse combination of a few active concepts.

In this idealized setting, concept directions are assumed to form a structured basis of the latent space, in which each concept is independently controllable.

When the representational capacity of the model is limited, such a decomposition leads to interference between concepte. In this regime, multiple semantic concepte are encoded within shared directions, a phenomenon known as superposition. This can be interpreted as a consequence of compression pressure, where the model prioritizes efficient encoding over strict factorization of concepte.

This perspective is closely related to recent work in mechanistic interpretability, where neural representations are assumed to decompose into sparse, approximately linear concepte. It also connects naturally to classical results in dictionary learning and sparse coding, where signals are represented as sparse combinations of elements from an overcomplete basis.

Sparse autoencoders instantiate this idea in practice by learning an overcomplete dictionary of concepte while enforcing sparsity constraints on activations. From this perspective, they can be interpreted as a tool for recovering a more disentangled basis of concepte from a superposed representation space.

Despite their empirical success, sparse autoencoders rely on strong structural assumptions that are not always justified. First, the learned concepte are not guaranteed to be identifiable: different training runs may yield different decompositions of the same latent space, even when the reconstruction performance is similar. Second, the assumption that the underlying representation is approximately linearly decomposable may not hold uniformly across all regions of the latent space, especially in highly nonlinear regimes. Finally, sparsity itself introduces a modeling bias that may favor certain types of concepte while suppressing others.

These limitations suggest that sparse autoencoders, while powerful, do not fully resolve the problem of entanglement in neural representations. They instead provide one possible coordinate system in which parts of the structure become more explicit, but not necessarily a complete or canonical one.

From this perspective, both interpretable directions and sparse autoencoders rely on a shared underlying assumption: that semantic structure can be recovered

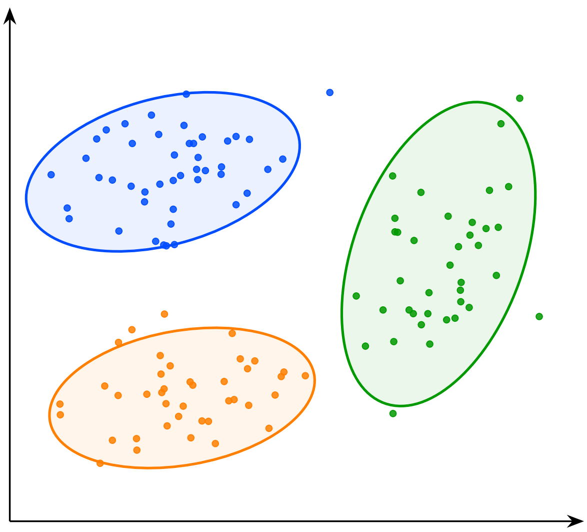


Figure 1.5: Visualisation de la structure de l'espace latent. Les points représentent les embeddings associés aux concepts A, B et C. Les ellipses indiquent l'étendue approximative de chaque groupe dans l'espace latent et mettent en évidence leur séparation géométrique.

through linear or approximately linear decompositions. While this assumption has led to significant empirical progress, it remains theoretically fragile.

The following section therefore introduces a complementary approach that relaxes this constraint and does not require a globally linear feature decomposition of the latent space.

1.5 Clustering

The latent space analysis methods presented so far all rely, either explicitly or implicitly, on assumptions about the linear organization of concepts within the latent space. For probing methods, concepts must be at least partially linearly separable for simple probes to succeed. Interpretability through latent directions assumes that variations associated with a concept can be captured by moving along a linear axis of the latent space. Similarly, the superposition hypothesis assumes that concepts are represented through linear directions that become entangled because of representational capacity constraints. Although these assumptions have led to numerous empirical successes, their theoretical justification remains limited. More importantly, several observations suggest that the linearity hypothesis may not fully capture the structure of latent representations.

For instance, some concepts can only be recovered using non-linear probes. Likewise, methods based on interpretable directions typically identify only a limited number of semantic factors. Finally, in sparse autoencoder approaches, enforcing excessive sparsity often degrades reconstruction quality and interpretability, suggesting that the underlying representation may not be perfectly described by a sparse linear decomposition.

Taken together, these observations indicate that semantic concepts may not always be organized according to a simple linear structure. This motivates the exploration of alternative approaches that rely on weaker assumptions regarding the geometry of latent spaces. One such approach is clustering-based analysis. Rather than assuming that concepts correspond to directions or linear subspaces, clustering methods only assume that semantically similar inputs produce nearby latent representations. Conversely, inputs that differ significantly from a semantic perspective are expected to be located farther apart in the latent space.

Under this assumption, the discovery of concepts can be reformulated as the identification of groups of embeddings that occupy the same region of the latent space. Concepts are therefore no longer associated with directions, but with clusters of neighboring representations. This idea forms the basis of the research field known as deep clustering. In these approaches, a collection of inputs is first projected into a latent space using a neural network. The resulting embeddings form a point cloud on which standard clustering algorithms can be applied. The obtained clusters are then interpreted as representing distinct semantic concepts.

This methodology enables the discovery of semantic structure in the latent space without requiring concepts to be linearly encoded. As a result, it can reveal forms of organization that would remain inaccessible to methods based solely on linear separability or interpretable directions. Deep clustering has been particularly successful in unsupervised image classification. Typically, an autoencoder is trained to learn a compact latent representation that preserves the essential information contained in the input data. Clustering algorithms are then applied to the learned embeddings. Empirically, the resulting clusters often correspond to meaningful semantic categories, despite the absence of supervision during training.

More advanced approaches combine representation learning and clustering objectives within a single optimization process. In this setting, additional loss terms encourage embeddings to organize themselves around cluster centroids during training. This often produces latent spaces that are easier to analyze and improves the quality of the discovered semantic structure.

Despite its effectiveness, clustering-based analysis also presents important limitations. First, as with most unsupervised semantic discovery techniques, the interpretation of the resulting clusters remains partially subjective. While benchmark datasets often provide class labels that facilitate evaluation, there is generally no guarantee that the discovered clusters correspond to the categories that humans consider most relevant. Alternative semantic groupings may be equally valid.

Second, clustering methods are highly sensitive to design choices. The number of clusters, the distance metric, and the clustering algorithm itself can significantly influence the results. Selecting these parameters is often challenging and may require substantial empirical tuning. Finally, semantic concepts do not necessarily form well-separated clusters in latent space. Some concepts may overlap, exhibit hierarchical relationships, or vary continuously rather than discretely. In such cases, a clustering-based description may provide only a partial view of the underlying representational structure.